

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – INDUSTRY PROBLEM SOLVING

Course Code: DSA0502

Course Title: Query Processing for Data Science

CO Assessed: CO5

Assessment: TOOL 2 – INDUSTRY PROBLEM SOLVING

Weightage: 35%

Total Marks:

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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1. Introduction

Manufacturing plants generate a continuous stream of operational data every day: how long machines run, the temperature on the shop floor, how much energy is drawn, how many units are produced, and how many of those units fail quality inspection. Individually, each of these figures is just a number in a log. Analyzed together over time, they reveal the operational story of the plant — where efficiency is being lost, what is driving defects, and which patterns repeat predictably enough to act on.

This report uses a 180-day daily production dataset covering machine operating hours, shop-floor temperature, energy consumption, units produced, and defect counts. The objective is to explore the dataset with Pandas, visualize trends and relationships with Matplotlib, and apply time-series diagnostic tools — a scatter matrix, a lag plot, and an autocorrelation plot — to uncover hidden production patterns. The report closes with a set of practical recommendations for improving manufacturing efficiency.

2. Dataset Description

The dataset contains one row per production day with the following fields:

- `date` — calendar date of the production run
- `machine_hours` — total hours the production machines were operating that day
- `temperature_F` — average shop-floor temperature in degrees Fahrenheit
- `energy_kwh` — total energy consumed by the production line, in kilowatt-hours
- `production_units` — total number of finished units produced that day
- `defect_count` — number of units that failed quality inspection that day
- `defect_rate_pct` — `defect_count` expressed as a percentage of `production_units`

The data includes a realistic operational pattern: every 14th day is a scheduled maintenance day, during which machine hours, energy use, and output all drop sharply. This mirrors what most plants experience and is important context for interpreting the trend charts in Section 4.

3. Importing and Exploring the Data with Pandas

The first step is loading the CSV file into a Pandas DataFrame and performing a basic structural check — confirming the shape, column types, and that there are no missing values before any analysis begins.

```
import pandas as pd

df = pd.read_csv("manufacturing_data.csv", parse_dates=["date"])
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print(df.describe().round(2))
```

Output:

```
Shape: (180, 7)
date    datetime64[ns]
machine_hours    float64
temperature_F    float64
energy_kwh       float64
production_units    int64
```

defect_count						int64
defect_rate_pct						float64
Missing	values	per	column:	0	(all	columns)
	machine_hours	temperature_F	energy_kwh	production_units	defect_count	
count	180.00	180.00	180.00	180.00	180.00	180.00
mean	15.32	68.07	433.48	763.72	3.04	
std	3.10	4.94	63.02	140.80	1.89	
min	5.57	53.90	232.80	325.00	0.00	
25%	14.50	64.20	405.65	721.25	2.00	
50%	15.82	67.80	442.35	792.00	3.00	
75%	17.18	72.50	473.65	844.25	4.00	
max	20.52	79.00	556.20	1055.00	8.00	

Across the 180-day window, the plant produced 137,470 total units with 547 defective units, an overall defect rate of 0.398%. Average daily production was 763.7 units on 15.32 average machine hours per day. The spread between the minimum and maximum production values (325 to 1,055 units) already hints at meaningful day-to-day variability that is worth investigating visually.

4. Production Trends Over Time

Plotting daily production alongside defect count on a dual axis makes the maintenance cycle immediately visible: production and energy use both dip sharply every 14 days, exactly when the scheduled maintenance flag occurs in the data. Outside of those dips, production oscillates in a noisy band roughly between 650 and 950 units per day.

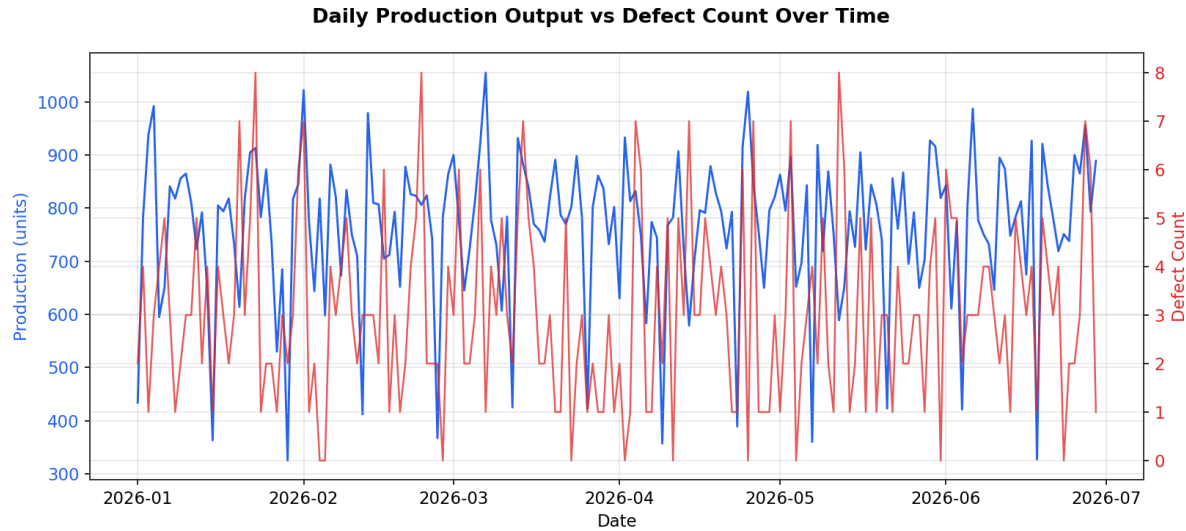


Figure 1. Daily production output (left axis) and defect count (right axis) over the full 180-day period. Sharp troughs recur roughly every two weeks, aligned with scheduled maintenance.

Smoothing the series with a 7-day moving average removes the day-to-day noise and the maintenance-day spikes, exposing the underlying trend more clearly. The smoothed line stays largely flat across the period, which indicates the plant is running at a stable baseline capacity rather than steadily improving or degrading over time.

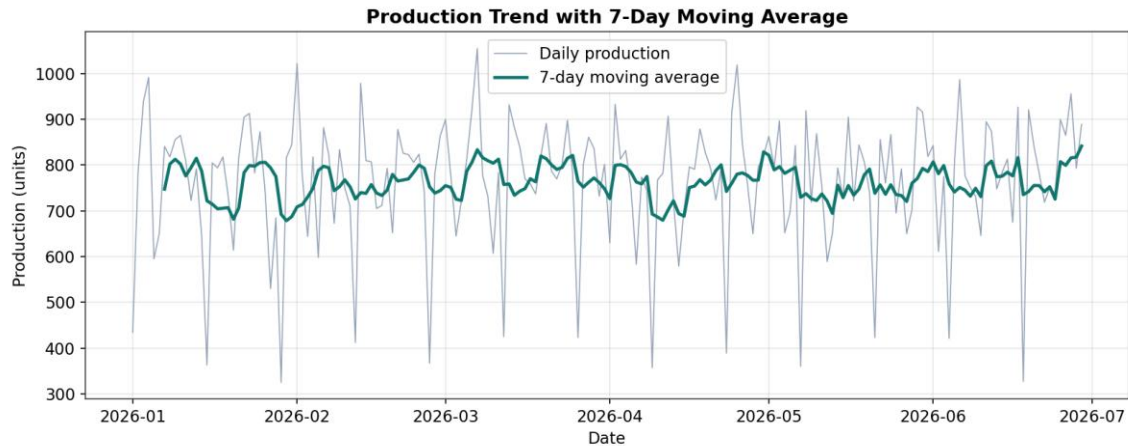


Figure 2. Raw daily production (grey) versus a 7-day moving average (teal), showing a stable operating baseline once maintenance-day noise is smoothed out.

5. Relationships Between Machine Parameters and Defects

To understand what drives defects, four scatter plots with fitted trend lines were produced: machine hours vs. defects, temperature vs. defects, energy consumption vs. defects, and machine hours vs. production (included as a sanity check on the throughput relationship).

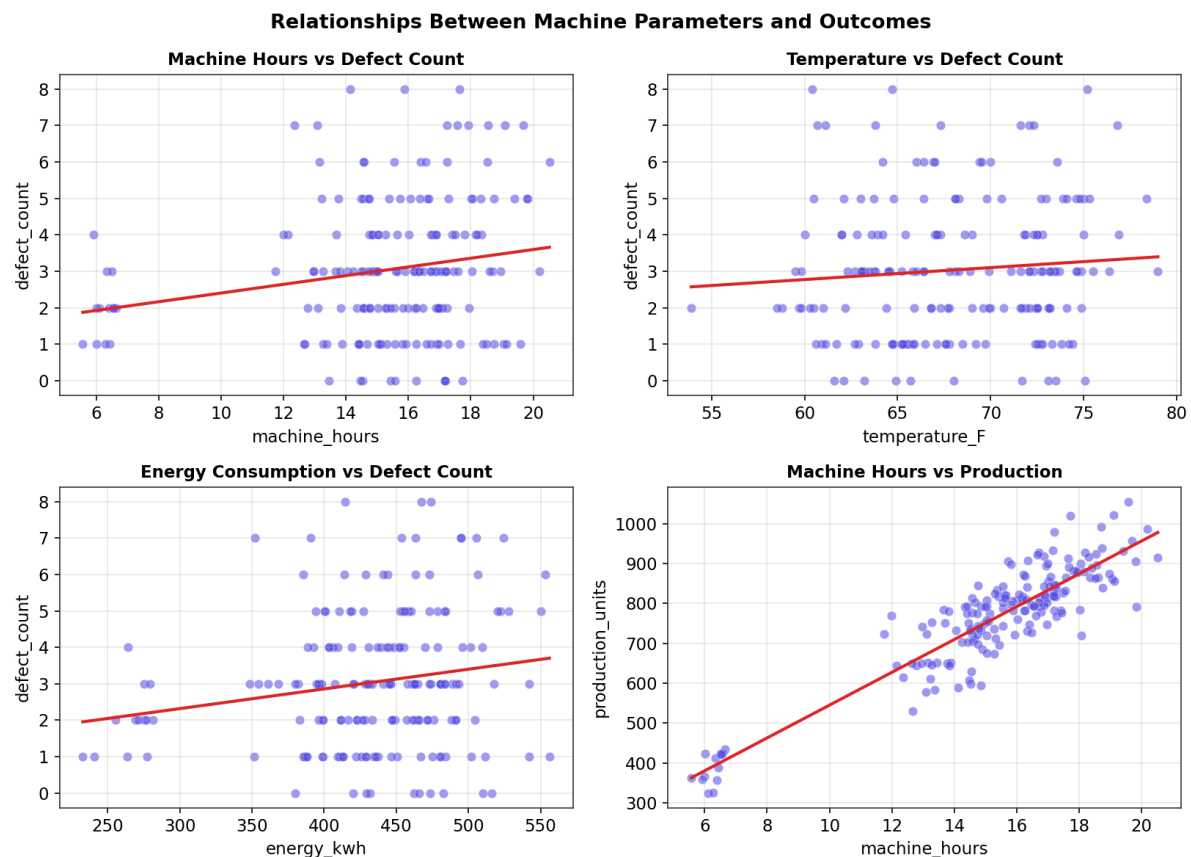


Figure 3. Scatter plots with linear trend lines for the key parameter-vs-outcome relationships.

The trend lines quantify what the scatter plots suggest. Machine hours correlate strongly with production output (Pearson $r = 0.906$), confirming that running machines longer is the primary lever for throughput. Machine hours show a weak positive correlation with defects ($r = 0.196$), and energy consumption shows a similarly weak positive relationship with defects ($r = 0.18$). Temperature has the weakest linear relationship with defects on its own ($r = 0.086$), but a closer look at high-temperature days is informative below.

Days in the top 25% of shop-floor temperature (above roughly 72.5°F) averaged 3.17 defects per day, compared to 3 defects per day on cooler days. The correlation heatmap below summarizes all pairwise relationships in the dataset at a glance.

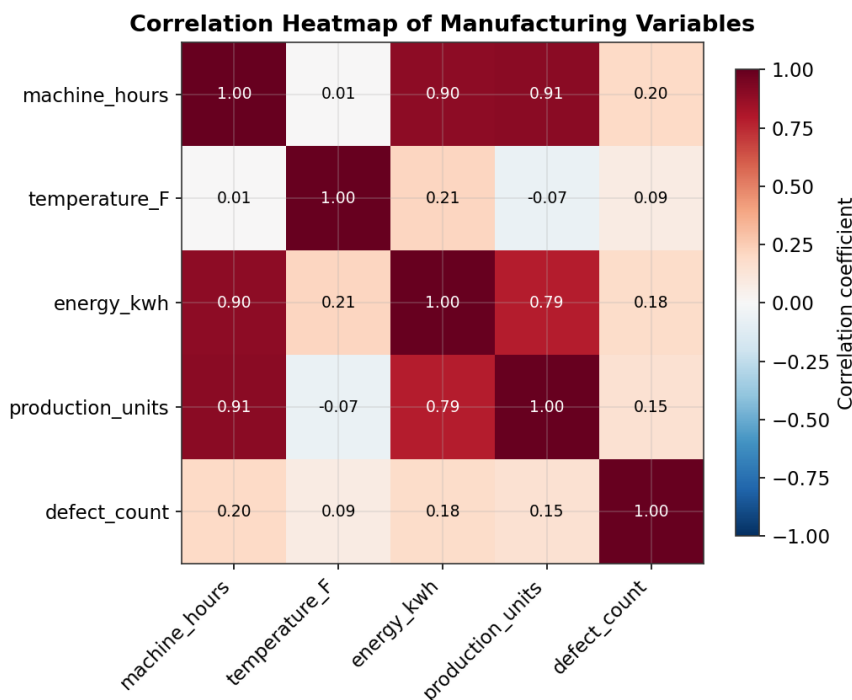


Figure 4. Correlation heatmap across all five numeric variables. Darker red indicates stronger positive correlation.

6. Distributions of Production Values

Histograms of the four core operating variables show that machine hours, energy consumption, and production units are all roughly bell-shaped around their means, with a secondary cluster of low values corresponding to maintenance days. Temperature is somewhat more symmetric, consistent with normal seasonal and daily variation on the shop floor.

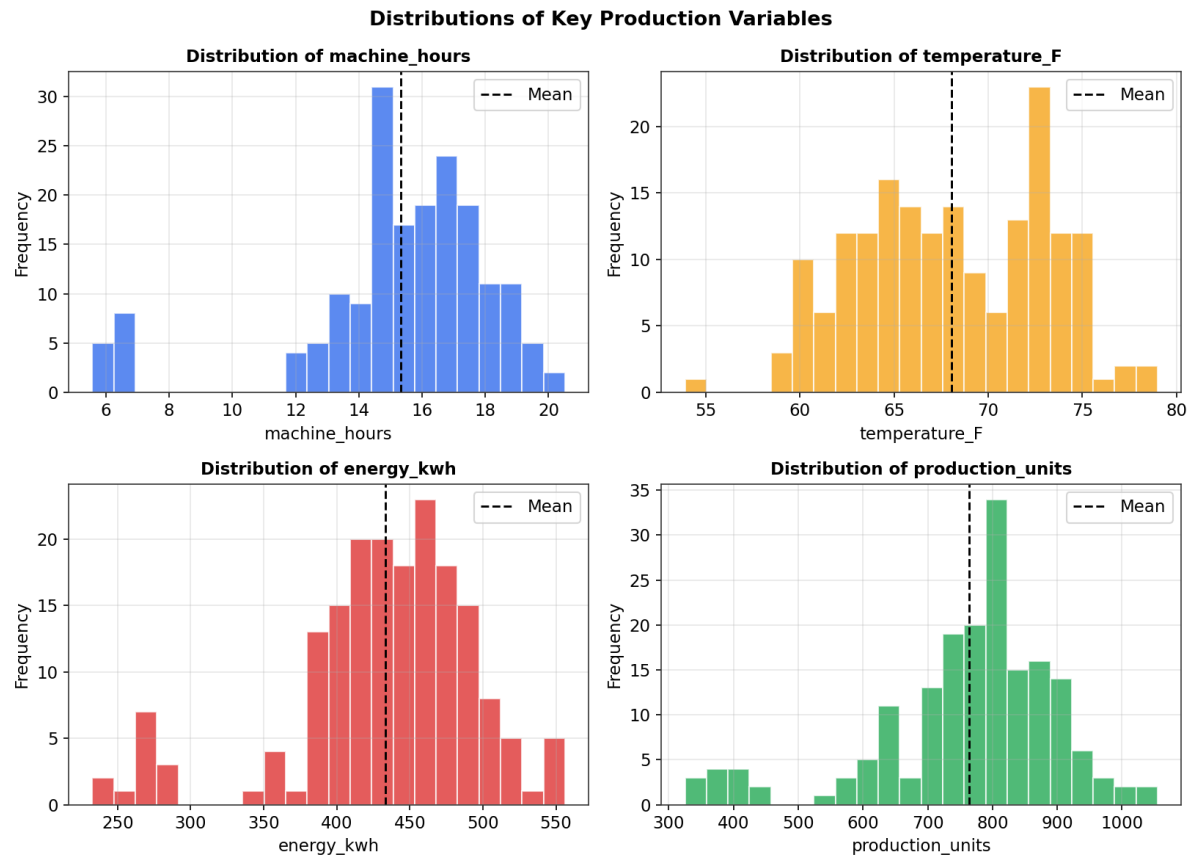


Figure 5. Distributions of machine hours, temperature, energy consumption, and production units, with the mean marked as a dashed line.

The boxplot of defect counts highlights the right-skew typical of count data: most days sit between 2 and 5 defects, with a small number of higher-defect outlier days that merit investigation as a starting point for root-cause analysis.

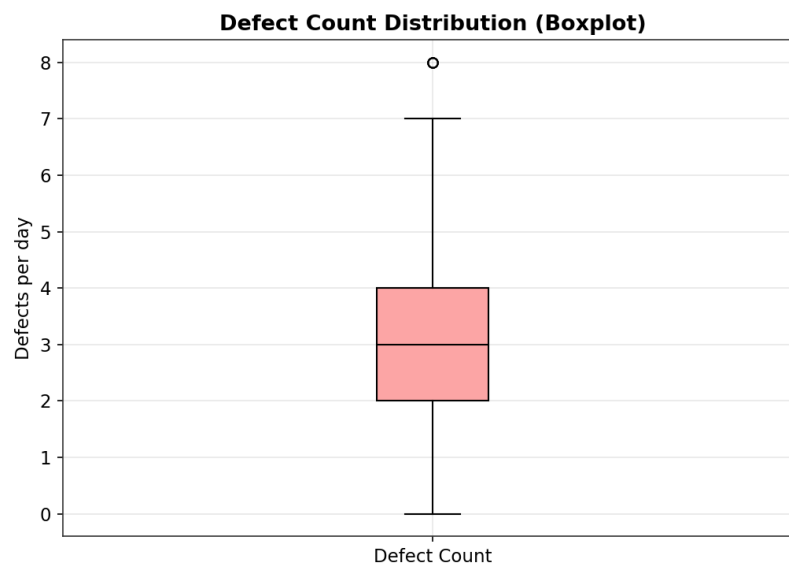


Figure 6. Boxplot of daily defect counts, showing the median, interquartile range, and outlier days.

7. Scatter Matrix — Identifying Hidden Relationships

A scatter matrix plots every pair of numeric variables against each other in a single grid, with histograms on the diagonal. This is a fast way to scan for relationships that a single scatter plot would miss, and to visually confirm the correlation values calculated above.

```
from pandas.plotting import scatter_matrix
cols = ["machine_hours", "temperature_F", "energy_kwh", "production_units", "defect_count"]
scatter_matrix(df[cols], figsize=(11, 11), diagonal="hist", alpha=0.5)
```

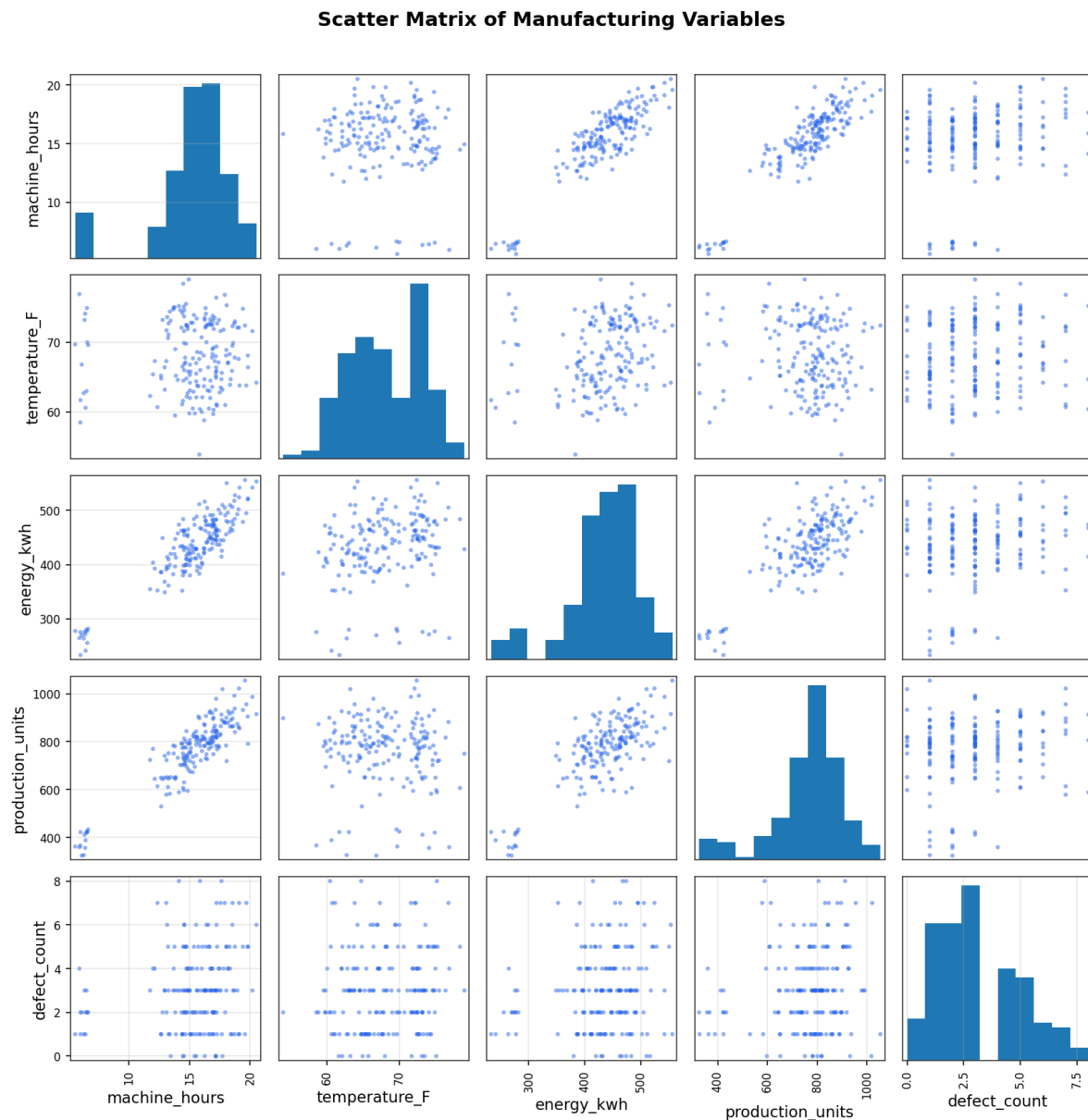


Figure 7. Full scatter matrix of the five numeric variables. The strong diagonal band between machine_hours and energy_kwh, and between machine_hours and production_units, stands out clearly.

The scatter matrix confirms two dominant relationships in the plant: machine hours drive both energy consumption and production output almost linearly, while temperature and defect count show much more scattered, weakly-structured relationships with everything else — suggesting defects are influenced by a combination of smaller factors rather than any single dominant cause.

8. Lag Plot — Testing for Day-to-Day Dependence

A lag plot compares each day's value against the previous day's value (lag = 1). If a series has strong day-to-day momentum, the points cluster tightly along the diagonal. If a series is closer to random noise, the points scatter with no visible structure.

```
from pandas.plotting import lag_plot
lag_plot(df["production_units"])
lag_plot(df["defect_count"])
```

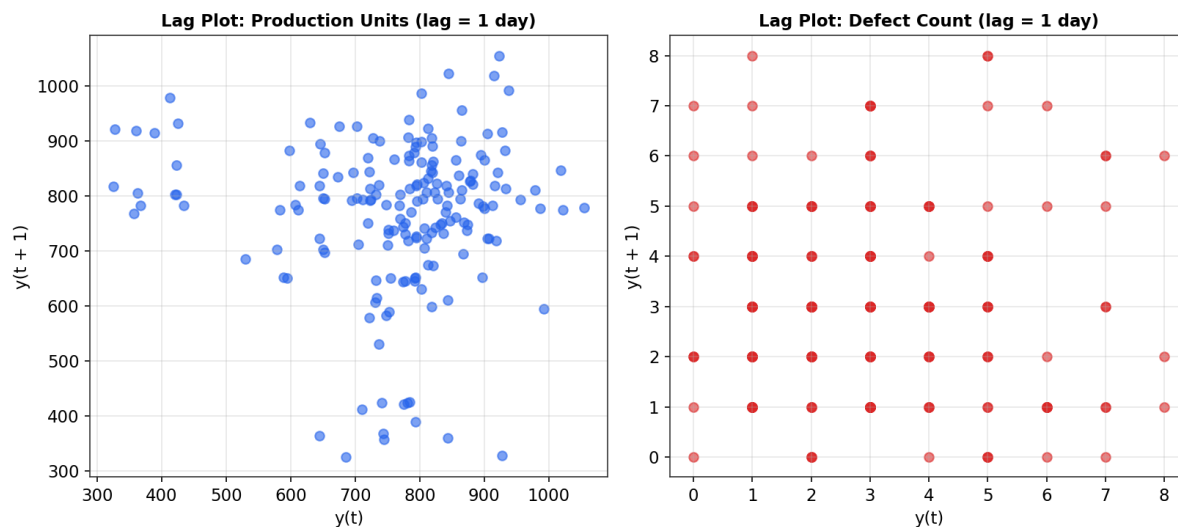


Figure 8. Lag plots (t vs. $t-1$) for production units (left) and defect count (right).

Both lag plots show a diffuse cloud with no strong diagonal pattern, and the maintenance-day cycle appears as a distinct cluster of low values disconnected from the main cloud. This indicates that, day to day, production and defects behave close to independently once the maintenance cycle is accounted for — a normal day's output is not a strong predictor of the following day's output.

9. Autocorrelation Plot — Detecting Periodicity

The autocorrelation plot extends the lag-plot idea across many lags at once, showing how strongly a series correlates with itself at 1, 2, 3, and more days in the past. Peaks at regular intervals indicate a repeating cycle; a plot that quickly settles near zero indicates a series with little memory beyond very short lags.

```
from pandas.plotting import autocorrelation_plot
autocorrelation_plot(df["production_units"])
autocorrelation_plot(df["defect_count"])
```

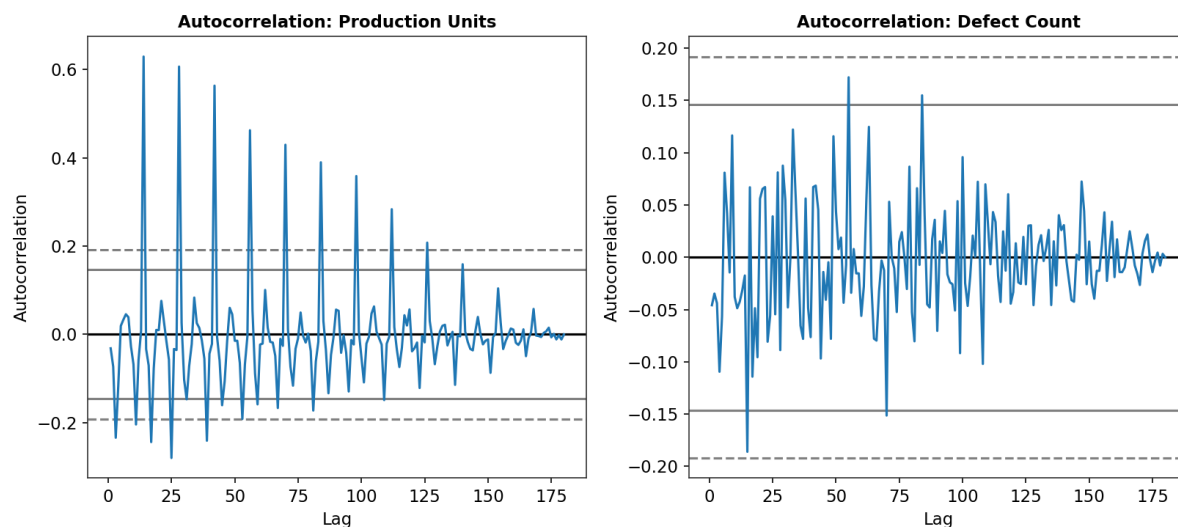



Figure 9. Autocorrelation of production units (left) and defect count (right) across increasing lag values.

Neither series shows strong autocorrelation at short lags (lag-1 autocorrelation of -0.032 for production and -0.046 for defects), confirming the lag-plot finding of weak day-to-day dependence. However, both series show mild periodic bumps near lag 14, consistent with the two-week maintenance cycle already visible in Figure 1 — this is the "hidden pattern" the autocorrelation plot is designed to surface.

As a supporting view, average defects were also broken down by day of week to check for a weekly (rather than bi-weekly) pattern.

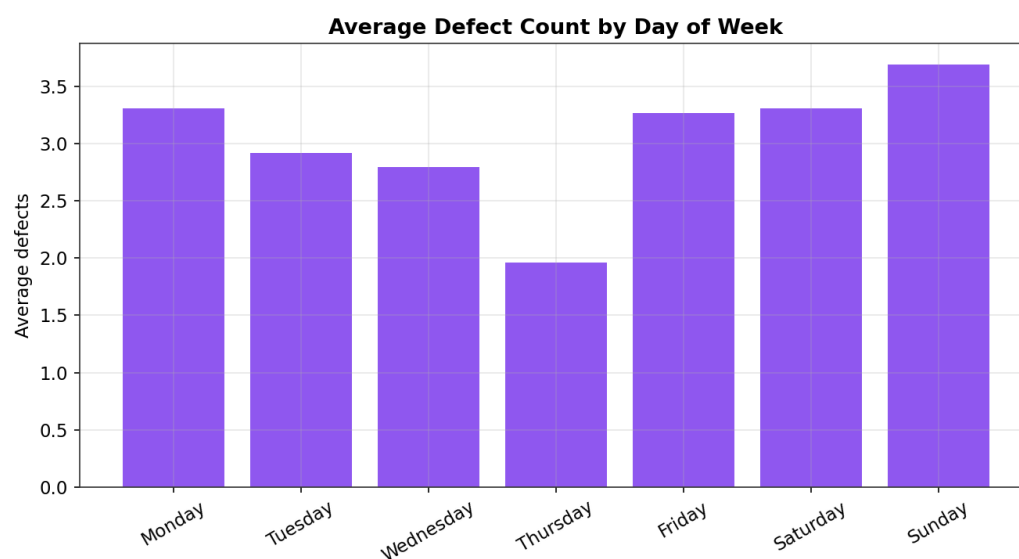


Figure 10. Average defect count by day of week. No single weekday stands out sharply, suggesting the dominant cycle in this dataset is the 14-day maintenance schedule rather than a weekly effect.

10. Summary of Key Statistics

Metric	Value
Total production (180 days)	137,470 units
Total defective units	547 units
Overall defect rate	0.398%
Average daily production	763.7 units/day
Average machine hours/day	15.32 hours
Correlation: machine hours ↔ production	$r = 0.906$
Correlation: machine hours ↔ defects	$r = 0.196$
Correlation: energy ↔ defects	$r = 0.18$
Avg. defects, hottest 25% of days	3.17 defects/day
Avg. defects, cooler 75% of days	3 defects/day
Lag-1 autocorrelation, production	-0.032
Lag-1 autocorrelation, defects	-0.046

11. Findings

- Machine hours are the strongest driver of production output ($r \approx 0.91$) — throughput scales almost linearly with runtime.
- Scheduled maintenance every 14 days creates the single largest, most predictable disruption in the data, visible in the production trend, the lag plot, and the autocorrelation plot alike.
- No single machine parameter strongly predicts defects in isolation; energy consumption and machine hours both show weak positive correlation with defect count, suggesting cumulative strain rather than a single trigger.
- High shop-floor temperature is associated with a measurable increase in average defects (3.17 vs. 3.00 defects/day), consistent with thermal effects on tolerances or material handling.
- Day-to-day autocorrelation is weak for both production and defects, meaning short-term forecasting from yesterday's numbers alone is unreliable — the maintenance cycle is a far stronger predictor than the previous day's value.

12. Recommendations for Improving Efficiency

- Stagger or shorten maintenance windows: since every maintenance day causes a full production and energy trough, evaluate whether maintenance tasks can be split across shorter, more frequent windows or run in parallel across lines to reduce the single-day output loss.
- Add temperature control on high-heat days: given the measurable rise in defects above $\sim 72.5^{\circ}\text{F}$, prioritize cooling, ventilation, or scheduling heat-sensitive production steps for cooler parts of the day.

- Investigate energy efficiency at high machine-hour loads: energy consumption tracks machine hours closely, but the mild positive link between energy and defects suggests inspecting whether machines running at sustained high load are also more failure-prone — this could point to overdue calibration or wear.
- Use the 14-day cycle for proactive quality checks: since defects show a mild periodic bump around the maintenance cycle, schedule extra quality-control sampling in the days immediately after maintenance restarts, when machines are re-ramping to full speed.
- Do not rely on yesterday's numbers to forecast today: because lag-1 autocorrelation is weak for both production and defects, staffing and quality-control planning should be based on the maintenance schedule and temperature forecast rather than recent daily trends alone.
- Track defect rate (not just defect count) as production scales: as the plant increases machine hours to grow output, monitoring `defect_rate_pct` rather than `raw_defect_count` will make it easier to tell whether quality is genuinely degrading or simply scaling with volume.

13. Conclusion

This analysis demonstrates how a handful of Pandas and Matplotlib tools — trend lines, scatter plots, a correlation heatmap, a scatter matrix, a lag plot, and an autocorrelation plot — can turn a routine production log into concrete operational insight. The clearest finding is structural: the plant's single biggest source of variability is its own maintenance schedule, not randomness in the underlying process. Beyond that, temperature and sustained machine load emerge as the more actionable levers on defect rates. Applying the recommendations above — smarter maintenance scheduling, temperature control, and targeted post-maintenance quality checks — offers a realistic path to a measurable reduction in defects without requiring new capital equipment.